



RUWANI VIDANACHCHI

Software Engineer

Oulu, Finland

Email: ruwanividanachchi@gmail.com | Phone: +358 417 270 289

LinkedIn: www.linkedin.com/in/ruwani-vidanachchi-815ab0294

GitHub: <https://github.com/ruwanividanachchi>

Portfolio : <https://my-portfolio-gamma-five-20.vercel.app>

SUMMARY

Bachelor of Engineering (Information Technology) graduate from Oulu University of Applied Sciences with strong software engineering skills in backend development, web technologies, cloud services, and mobile applications. Experienced in designing and building full-stack applications using Java, Python, TypeScript, React, Next.js, Kotlin, PostgreSQL, and REST APIs. Passionate about clean code, scalable software, cloud technologies, AI-assisted applications, and solving real-world engineering problems. Actively seeking Graduate Software Engineer and Full-Stack Developer opportunities.

EDUCATION

Bachelor of Engineering (BEng) in Information Technology

Oulu University of Applied Sciences, Finland

Aug 2023 – May 2026

GPA: 4.37 / 5.0

Specialization: Web Development

Relevant studies: Software Engineering, Mobile Development, Cloud Services, IoT Applications, Data Storage & Analysis, Operating Systems, Advanced Web Applications, Applied Mathematics & Physics

PROJECT EXPERIENCE

Interactive Building Modelling and Heat Load Analysis Platform (Bachelor's Thesis)

March 2026 – May 2026

Next.js | TypeScript | Three.js | Supabase | REST APIs

GitHub Link

- Developed a full-stack web application for interactive building modelling and HVAC cooling load analysis.
- Built 2D floor-plan editing and 3D building visualization using Three.js with real-time engineering calculations.
- Integrated external APIs and Supabase for data management, authentication, and project storage.
- Applied modular software engineering practices to develop a scalable and user-friendly platform.

AI-Assisted Job Application Tracker – Web Application (Software R&D Project)

Jan 2026 – Present

Next.js (App Router) | TypeScript | Gemini API | Tailwind CSS | Recharts | React Context

GitHub Link

- Designed and implemented an AI-assisted web application for tracking job applications and analyzing job-resume matching
- Developed modular frontend and server-side logic using Next.js App Router, including authentication flows, dashboard views, analytics components, and state management
- Integrated Google Gemini API to extract skills from job descriptions, compute match scores, identify skill gaps, and generate improvement suggestions
- Focused on code reliability, modular design, and testable components to support future extension

Automated Distribution Warehouse – Robotics & Systems Project

Sep 2025 – Dec 2025

Python | ROS | Universal Robot Arm | GoPiGo | Tello EDU Drone | Web Integration

GitHub Link

- Designed and implemented a distributed warehouse simulation where drones, cobots, and mobile robots collaborated via software-controlled workflows to fulfill web-based orders
- Developed multi-device communication and automation logic, integrating robotics control, system coordination, and networked processes
- Gained hands-on experience in robotics programming, system integration, and real-world automation concepts through implementation and industry exposure

AstroTrack – Android Application (Software R&D Project)

Mar 2025 – Apr 2025

Kotlin | Jetpack Compose | Firebase | REST API | MVVM

GitHub Link

- Developed a modular native Android application integrating external REST APIs, secure authentication, and real-time cloud data
- Implemented background processing and scheduled tasks using WorkManager, focusing on reliability and maintainability
- Designed a responsive UI using Jetpack Compose and MVVM architecture, enabling scalable development and clear separation of concerns

OAMKINO – Full-Stack Software Engineering Project

Oct 2024 – Dec 2024

React | Node.js | Express | PostgreSQL | Finnkino API | JWT

GitHub Links: Client | Server

- Developed a full-stack software system with secure authentication and modular RESTful backend services
- Integrated external APIs and relational databases, ensuring data integrity and reliable backend logic
- Implemented access control, error handling, and API security following software engineering best practices

Travel Blog Website – Web Software Project

Mar 2024 – May 2024

HTML | CSS | JavaScript | Node.js | Express.js | PostgreSQL

GitHub Link

- Developed a database-driven web application with user authentication and full content management functionality
- Designed and implemented backend logic, data models, and API endpoints for reliable data handling
- Applied Agile/Scrum practices, version control, and responsive design principles

TECHNICAL SKILLS

- Programming Languages – Python, Java, Kotlin, JavaScript, TypeScript, C (basic)
- Frameworks & Technologies – Next.js, React, Node.js, REST APIs, Android (Jetpack Compose), React Native
- Software Engineering & Backend – Clean and maintainable code, testing & debugging, modular architecture, Agile/Scrum collaboration, technical documentation
- Systems & Tools – Linux, Git, operating systems fundamentals, cloud services
- AI & Data (Exposure) – AI-assisted workflows, data analysis logic, analytics and visualization
- Databases & Data – PostgreSQL, MySQL, data modeling
- Cloud & Development Tools – Firebase, Supabase, VS Code, Android Studio
- Robotics & Automation – ROS, Python scripting, UR robots, GoPiGo, Tello EDU

LANGUAGES

- English (Fluent) Finnish (Intermediate) Sinhalese (Native Speaker)

EXTRACURRICULAR ACTIVITIES

International Tutor 2025-26

Oulu University of Applied Sciences, Oulu, Finland

- Supported incoming international students with orientation, integration, and academic guidance.
- Facilitated intercultural communication and helped students adapt to Finnish student life.
- Organized social and cultural activities to promote community building and well-being.

CERTIFICATIONS

Introducing Generative AI with AWS

Verified certificate

Udacity (Part of Accenture) – June 29, 2025